Thompson's group F is nonamenable

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Abstract

We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture and resolves the amenability problem for F. The proof is one finite averaging argument: a Følner set for a certain fixed finite subset of F would produce an approximate fixed point of a Lipschitz self-map of the Hilbert unit ball that has none.

1 The result and how it is proved

1.1 Statement

Thompson's group F is the set of homeomorphisms $g : [0,1] \to [0,1]$ that are increasing and piecewise linear with finitely many pieces, whose breakpoints are dyadic rationals (numbers $k2^{-r}$ with k, r integers), and whose slopes lie in $2^{\mathbb{Z}} = \{2^{\lambda} : \lambda \in \mathbb{Z}\}$. The product is composition, $hg = h \circ g$, and F is regarded as a discrete group.

A discrete group G is *amenable* if there is a linear functional $\mu : \ell^{\infty}(G;\mathbb{R}) \to \mathbb{R}$ with $\mu(1) = 1$, $\mu(\psi) \geq 0$ whenever $\psi \geq 0$, and $\mu(g \mapsto \psi(hg)) = \mu(\psi)$ for every $h \in G$. This invariant-mean viewpoint belongs to the classical theory of von Neumann and Day [19, 6]. The only fact about amenability used in this paper is the forward implication of the Følner criterion [7], which we take as a cited result:

> *If G is amenable, then for every finite $S \subset G$ and every $\varepsilon > 0$ there is a nonempty finite $A \subset G$ such that*
>
> $$\frac{|hA \triangle A|}{|A|} < \varepsilon \qquad \text{for every } h \in S. \tag{1.1}$$

Here $hA = \{ha : a \in A\}$ and $\triangle$ is symmetric difference. The source takes (1.1) from the left-translation formulation in [14, Theorem 1.2]. The source does not quote that theorem; it describes it as giving, for every finite S and every tolerance, a nonempty finite A for which the *sum* over $h \in S$ of the ratios $|hA \triangle A|/|A|$ is bounded by the tolerance. Each ratio is nonnegative, hence at most the sum, so applying that formulation with tolerance $\varepsilon/2$ gives the strict bound (1.1) for each h separately, whether the cited bound is strict or not.

Theorem 1.1. *Thompson's group F is not amenable.*

What is actually proved is quantitative (Proposition 2.6): there are a finite set $S \subset F$ and a constant $\eta_0 > 0$ such that $\max_{h \in S} |hA \triangle A|/|A| \geq \eta_0$ for *every* nonempty finite $A \subset F$. This contradicts (1.1) with $\varepsilon = \eta_0$.

1.2 The one analytic input

Lemma 1.2 (Benyamini–Sternfeld)**.** *There are a real Hilbert space H, a Lipschitz map $f : B \to B$ on its closed unit ball B, and a constant $\delta > 0$ such that*

$$\|f(x) - x\| \geq \delta \qquad (x \in B).$$

Benyamini and Sternfeld prove this for every infinite-dimensional normed space [2, Theorem, part (3), p. 439]. Appendix A gives a self-contained construction on $L^2([0,1];\mathbb{R}^2)$ with $\delta = 1/2$. The lemma is used in Section 2.2 to define the coloring and in Step 4 of the proof of Proposition 2.6. No such f exists when $\dim H < \infty$, by Brouwer's fixed-point theorem, so infinite dimension is essential.

1.3 Proof at a glance

Objects. Fix f, L, δ as in Lemma 1.2 (L a Lipschitz constant of f) and an integer $D > 4L^2/\delta^2$. A *basic interval* is a dyadic interval $[k2^{-r}, (k+1)2^{-r}] \subseteq [0,1]$. For a basic interval I, $s_I : [0,1] \to I$ is the increasing affine bijection.

- *Parents:* D basic intervals $I_1 < \cdots < I_D$ inside $(0,1)$ with positive gaps between consecutive ones.
- *Children:* the D^2 intervals $I_i \cdot I_j := s_{I_i}(I_j)$, that is, the same family $I_1 < \cdots < I_D$ copied affinely into each parent I_i. (The source calls them descendants.)
- *Colors:* to each finite partition T of $[0,1]$ into basic intervals we attach a vector $\mathrm{col}(T) \in B$ by recursion: if every parent is a union of cells of T, then $\mathrm{col}(T)$ is f applied to the average of the colors of the D partitions that T induces on the parents, each stretched back to $[0,1]$; otherwise $\mathrm{col}(T) = 0$.
- *Functions on F:* fix a level n and let $T^{(n)}$ be the uniform dyadic partition into 2^n cells. Call (n, g) *admissible* if the image partition $gT^{(n)}$ consists of basic intervals and every parent and every child is a union of its cells. Any finite set of elements g is admissible for all large n (Lemma 2.2). For admissible (n, g) and I a parent or a child, $X_I(g)$ is the color of $gT^{(n)}$ restricted to I and stretched back to $[0,1]$; for non-admissible (n, g) we set $X_I(g) = 0$, so that X_I is defined on all of F. *For admissible* (n, g) the recursion gives

$$X_{I_i}(g) = f(z_i(g)), \quad \text{where} \quad z_i(g) = \frac{1}{D}\sum_{j=1}^{D} X_{I_i \cdot I_j}(g), \qquad \text{and we put} \quad m(g) = \frac{1}{D}\sum_{i=1}^{D} X_{I_i}(g).$$

Key idea. Let $A \subset F$ be finite and think of g as uniformly random in A; $\mathbb{E}_A$ denotes the average over $g \in A$.

(i) *Group theory.* F contains, for each pair $I < J$ of disjoint parents or children with a positive gap, an element $h_{I,J}$ mapping I affinely onto I_1 and J affinely onto I_2. Colors are carried along exactly: $X_I(g) = X_{I_1}(h_{I,J}g)$ and $X_J(g) = X_{I_2}(h_{I,J}g)$. If A is almost invariant under the finite set S of these elements, with defect η, then the average over A of $\langle X_I, X_J\rangle$ is within η of one number α for all such pairs. The number α depends on A, but not on the pair.

(ii) *Hilbert space.* The quantity controlled is $\mathbb{E}_A\|z_i - m\|^2$. Expanding it into inner products $\langle X_I, X_J\rangle$, all but a fraction $1/D$ of the terms are pairs covered by (i); they contribute α with total coefficient $1 - 2 + 1 = 0$, up to errors η. The remaining terms are bounded by 1 in absolute value. The result is $\mathbb{E}_A\|z_i - m\|^2 \leq 4/D + 4(1 - 1/D)\eta$. (Heuristically, vectors whose pairwise correlations all equal α behave like one common vector plus uncorrelated noise; the proof does not use this picture, only the expansion just described.)

(iii) *Contradiction.* Since m is the average of the $f(z_i)$ and f is L-Lipschitz, (ii) gives $\mathbb{E}_A\|m - f(m)\|^2 \le L^2(4/D + 4(1-1/D)\eta)$. But $\|m - f(m)\| \ge \delta$ always. Hence η is bounded below by a positive constant independent of A.

Steps and dependencies.

Statement	Content	Used in
Lemma 1.2	the map f (proved in Appendix A)	definition of col; Step 4
Lemma 2.1	restriction is compatible with nesting	definition of col; Step 3
Lemma 2.2	$gT^{(n)}$ is basic and fine for large n	Step 0
Lemma 2.3	pair transport in F	definition of S
Lemma 2.4	transports preserve restricted partitions	Step 1
Lemma 2.5	averages over A and hA differ by the boundary	Step 2
Proposition 2.6	uniform boundary bound (Steps 0–7)	Theorem 1.1

Inside the proof of Proposition 2.6: Steps 1–2 give the correlation comparison (i); Step 3 gives $X_{I_i} = f(z_i)$; Step 4 is the pointwise inequality in (iii); Steps 5–6 are (ii); Step 7 combines them.

Order of choices. f, L, δ, D, the parents, the children and the finite set S are fixed once. Then A is arbitrary. Only the partition level n, and hence the number α, depend on A.

1.4 Context and prior work

Thompson introduced F in 1965; an early published construction appears in McKenzie and Thompson [11], with the identification explained by Cannon, Floyd, and Parry [5, pp. 215–216], who also record Geoghegan's 1979 conjecture that F is nonamenable [5, p. 227]. Two classical tests are silent: F contains no nonabelian free subgroup [3, Theorem 3.1], yet it is not elementary amenable [5, Theorem 4.10]. Brown and Geoghegan construct a classifying space for F with finitely many cells in each dimension [4]. Moore proves tower lower bounds on the sizes of Følner sets [15, Theorem 1.1] and characterizes amenability of F by a scalar convex Ramsey property for finite rooted ordered binary trees [14, Theorem 3.1]. Guba improves densities of finite Cayley subgraphs [8] and derives restrictions on right-invariant means [9]. Haagerup and Olesen prove that simplicity of the reduced C^*-algebra of Thompson's group T would imply nonamenability of F [10, Theorem 5.5]. There have been claims of a complete resolution in both directions: Akhmedov's 2021 version claims nonamenability through a height-function criterion [1]; Shavgulidze claimed amenability [18], and Moore identified errors in that approach [12]; Moore separately withdrew an amenability claim after Akhmedov identified an error in Lemma 4.13 of that manuscript [13] (a different work from the Ramsey characterization above). The source does not discuss how the present argument relates to those earlier claims, and none of the results in this subsection is used below. The proof is self-contained apart from the Følner implication (1.1): it uses only the standard dyadic-partition model of F [5, Section 2, Lemma 2.2], whose needed facts are proved in Section 2.1, and Lemma 1.2, proved in Appendix A. The claim of the abstract therefore stands or falls with Section 2 and Appendix A.

1.5 Notation changed from the source

The following symbols were renamed to avoid clashes. The color map p is now col, because p is a percolation parameter in Section 3. The invariant mean M is now μ. The finite set $K \subset F$ in

Lemma 2.2 is now Ω. The right-hand side of (2.8), unnamed in the source, is called η_0. In Section 3 the Hilbert space and operator of Corollary 3.1 are $\mathcal{H}$ and V, the generating set is Σ (the letter S is reserved for the transport set), the Cayley graph is $\mathrm{Cay}(F, \Sigma)$, and the connection kernel is $\mathcal{C}_p$. In Appendix A the map G is now Φ, the operator A is Λ, the quarter-turn J is written $\xi \mapsto \xi^{\perp}$, the second curve parameter s is τ, the constants K, M are K_γ, M_γ, and $\|e\|$ is written out instead of being abbreviated r. The symbol $\mathbb{E}_A$ (the source's) always means the average over $g \in A$, and the letter c is used only in Appendix A, for the constant $1/8$.

2 Proof of the theorem

2.1 Dyadic partitions, restriction, and transport

Definitions.

- A *basic interval* is $[k2^{-r}, (k+1)2^{-r}] \subseteq [0,1]$ with $r \in \mathbb{N} \cup \{0\}$ and $0 \le k < 2^r$; r is its *level* and 2^{-r} its length.
- A *basic partition* is a finite set T of basic intervals (its *cells*) with union $[0,1]$ and pairwise disjoint interiors. $|T|$ is the number of cells; the *mesh* is the greatest cell length.
- T *respects* an interval I if I is a union of cells of T.
- For intervals, $I < J$ means $\max I < \min J$ (a positive gap is required). Two intervals are *separated* if $I < J$ or $J < I$. An interval is *internal* if both endpoints lie in $(0,1)$.
- For a basic interval I, $s_I : [0,1] \to I$ is the increasing affine bijection, and $I \cdot J := s_I(J)$ for $J \subseteq [0,1]$.
- If T respects I, its *normalized restriction* is $T_I = \{s_I^{-1}(C) : C \in T,\ C \subseteq I\}$.
- $T^{(n)}$ is the partition of $[0,1]$ into the 2^n basic intervals of level n. For $g \in F$, $gT^{(n)} = \{g(C) : C \in T^{(n)}\}$.

Nesting fact. *If two basic intervals have intersecting interiors, one contains the other.* Let their levels be $r \le r'$. The endpoints of the level-r interval I are multiples of $2^{-r'}$, so neither lies in the interior of the level-r' interval I'. An open interval that meets the interior of I and contains no endpoint of I lies inside I; hence $I' \subseteq I$.

Lemma 2.1 (Restriction)**.** *Let I be a basic interval and T a basic partition respecting I.*

(a) If J is a basic interval, then $I \cdot J$ is a basic interval contained in I, and $s_{I\cdot J} = s_I \circ s_J$.

(b) T_I is a basic partition, and $|T_I|$ is the number of cells of T contained in I. If I is internal, then $|T_I| < |T|$.

(c) If T also respects $I \cdot J$ for a basic interval J, then T_I respects J and

$$(T_I)_J = T_{I\cdot J}. \tag{2.1}$$

This is used to see that the coloring recursion is well founded (b) and to derive the identity $X_{I_i} = f(z_i)$ in Step 3 below (c).

Proof. Write $I = [l2^{-r}, (l+1)2^{-r}]$, so $s_I(x) = 2^{-r}(x+l)$ and $s_I^{-1}(x) = 2^r x - l$.

(a) If $J = [k2^{-r'}, (k+1)2^{-r'}]$, then $s_I(J) = [(k + l2^{r'})2^{-(r+r')}, (k+1+l2^{r'})2^{-(r+r')}]$, a basic interval of level $r + r'$, contained in $s_I([0,1]) = I$. The map $s_I \circ s_J$ is an increasing affine bijection of $[0,1]$ onto $s_I(J) = I \cdot J$; such a bijection is unique, so it equals $s_{I\cdot J}$.

(b) Let $C = [k2^{-r'}, (k+1)2^{-r'}]$ be a cell of T with $C \subseteq I$. Its length is at most that of I, so $r' \geq r$, and

$$s_I^{-1}(C) = [(k - l2^{r'-r})2^{-(r'-r)},\ (k+1-l2^{r'-r})2^{-(r'-r)}] \subseteq [0,1]$$

is a basic interval of level $r' - r$. Since T respects I, the cells of T inside I have union I and disjoint interiors; the homeomorphism $s_I^{-1} : I \to [0,1]$ carries them to a basic partition, with one cell for each. If I is internal, the cell of T containing 0 is not inside I, so $|T_I| < |T|$.

(c) $I \cdot J$ is a union of cells of T. These cells lie in $I \cdot J \subseteq I$, so their images under s_I^{-1} are cells of T_I, with union $s_I^{-1}(I \cdot J) = J$. Thus T_I respects J. Using (a) in the form $s_J^{-1} \circ s_I^{-1} = s_{I \cdot J}^{-1}$ and $s_I^{-1}(C) \subseteq J \iff C \subseteq I \cdot J$,

$$(T_I)_J = \{s_J^{-1}(s_I^{-1}(C)) : C \in T,\ C \subseteq I,\ s_I^{-1}(C) \subseteq J\} = \{s_{I \cdot J}^{-1}(C) : C \in T,\ C \subseteq I \cdot J\} = T_{I \cdot J}. \quad \square$$

Lemma 2.2 (Fine images are basic)**.** *(a) For each $g \in F$ there is n_g such that $gT^{(n)}$ is a basic partition for all $n \geq n_g$; the mesh of $gT^{(n)}$ tends to 0 as $n \to \infty$.*

(b) Given a finite set $\Omega \subset F$ and a finite family $\mathcal{J}$ of basic intervals, there is n_0 such that $gT^{(n)}$ is basic and respects every member of $\mathcal{J}$ whenever $g \in \Omega$ and $n \geq n_0$.

Part (b) is used once, in Step 0 of the proof of Proposition 2.6.

Proof. (a) Let $0 = x_0 < x_1 < \cdots < x_M = 1$ be dyadic points such that $g(x) = 2^{\lambda_i} x + \nu_i$ on $[x_{i-1}, x_i]$, with $\lambda_i \in \mathbb{Z}$. Each $g(x_i)$ is dyadic, by induction from $g(x_0) = 0$ and $g(x_i) = g(x_{i-1}) + 2^{\lambda_i}(x_i - x_{i-1})$. Hence $\nu_i = g(x_{i-1}) - 2^{\lambda_i} x_{i-1}$ is dyadic. Choose an integer $N \geq 0$ such that all x_i and all ν_i are integer multiples of 2^{-N}, and put $n_g = N + \max_i |\lambda_i|$. Let $n \geq n_g$.

- Since $n \geq N$, every x_i is a multiple of 2^{-n}, so each cell $C = [k2^{-n}, (k+1)2^{-n}]$ of $T^{(n)}$ lies in one piece $[x_{i-1}, x_i]$.
- For such a cell, $g(C) = [k'2^{-(n-\lambda_i)}, (k'+1)2^{-(n-\lambda_i)}]$ with $k' = k + \nu_i 2^{n-\lambda_i}$. Here $n - \lambda_i \geq N \geq 0$, and k' is an integer because $\nu_i \in 2^{-N}\mathbb{Z}$ and $n - \lambda_i \geq N$. Since $g(C) \subseteq [0,1]$, it is a basic interval.
- g is an increasing homeomorphism of $[0,1]$, so the intervals $g(C)$ have union $[0,1]$ and disjoint interiors.

Thus $gT^{(n)}$ is a basic partition, of mesh at most $2^{\max_i \lambda_i - n} \to 0$.

(b) Let ℓ_0 be the smallest length of a member of $\mathcal{J}$. By (a), for each $g \in \Omega$ there is $n'_g \geq n_g$ with mesh of $gT^{(n)}$ less than ℓ_0 for $n \geq n'_g$; let $n_0 = \max_{g \in \Omega} n'_g$. Fix $g \in \Omega$, $n \geq n_0$, $I \in \mathcal{J}$, and put $T = gT^{(n)}$. Let x be an interior point of I and C a cell of T containing x. Since x lies in the closure of the interior of C, the interiors of C and I meet. By the nesting fact one contains the other, and C is shorter than I, so $C \subseteq I$. Hence the cells of T contained in I cover the interior of I, and, their union being closed, they cover I. So T respects I. $\square$

Lemma 2.3 (Pair transport)**.** *If $I < J$ and $I' < J'$ are pairs of internal basic intervals, there is $h \in F$ with $h(I) = I'$, $h(J) = J'$, and h affine on I and on J.*

This is the only place where the structure of F beyond Lemma 2.2 is used. It supplies the finite set S in Section 2.3.

Proof. Step 1 (gaps). The complement of $I \cup J$ in $[0,1]$ consists of three gaps: left of I, between I and J, and right of J. Their closures $\Gamma_0, \Gamma_1, \Gamma_2$ have positive length (internality and $I < J$) and dyadic endpoints. Define $\Gamma'_0, \Gamma'_1, \Gamma'_2$ likewise for $I' < J'$. Choose N so large that all endpoints of all these intervals are multiples of 2^{-N}. Then each Γ_i is a union of $p_i \geq 1$ basic intervals of level N, and each Γ'_i of $p'_i \geq 1$ such intervals.

Step 2 (equalize counts). Replacing a basic interval by its two halves gives two basic intervals with the same union and disjoint interiors, and raises the count by one. For each $i \in \{0,1,2\}$, perform $|p_i - p_i'|$ such bisections on whichever of Γ_i, Γ_i' has fewer cells; that side has at least one cell, so this is possible. Now Γ_i and Γ_i' are each divided into the same number P_i of basic intervals with disjoint interiors. Their levels may differ from cell to cell, which is harmless.

Step 3 (define h). Listing from left to right the P_0 cells of Γ_0, then I, then the P_1 cells of Γ_1, then J, then the P_2 cells of Γ_2 gives a basic partition $C_1, \dots, C_Q$ of $[0,1]$, with $Q = P_0 + P_1 + P_2 + 2$. The same recipe for the primed data gives $C_1', \dots, C_Q'$, with I' in the same position as I and J' in the same position as J. Let h map each C_k onto C_k' by the increasing affine bijection.

Step 4 ($h \in F$). The right endpoint of C_k is the left endpoint of C_{k+1}, and both definitions send it to the common endpoint of C_k' and C_{k+1}'; so h is a well-defined increasing homeomorphism of $[0,1]$. It is piecewise linear with at most Q pieces, its breakpoints are endpoints of basic intervals and hence dyadic, and its slope on C_k is $|C_k'|/|C_k|$, a ratio of two powers of 2 since every C_k and C_k' is basic. So $h \in F$, and by construction h maps I affinely onto I' and J affinely onto J'. □

Lemma 2.4 (Exact covariance)**.** *Let I, I' be basic intervals and $h \in F$ with $h(I) = I'$ and h affine on I. Let $g \in F$ and n be such that $gT^{(n)}$ and $(hg)T^{(n)}$ are basic partitions respecting I and I', respectively. Then, as sets of intervals,*

$$\left((hg)T^{(n)}\right)_{I'} = (gT^{(n)})_I. \tag{2.2}$$

This is used in Step 1 of the proof of Proposition 2.6.

Proof. The map $h \circ s_I$ is an increasing affine bijection of $[0,1]$ onto I', hence equals $s_{I'}$; equivalently $s_{I'}^{-1} \circ h = s_I^{-1}$ on I. Because $hg = h \circ g$, the cells of $(hg)T^{(n)}$ are exactly the sets $h(C)$ with C a cell of $gT^{(n)}$. Since h is a bijection with $h(I) = I'$, we have $h(C) \subseteq I' \iff C \subseteq I$. Therefore

$$\left((hg)T^{(n)}\right)_{I'} = \{s_{I'}^{-1}(h(C)) : C \in gT^{(n)},\ C \subseteq I\} = \{s_I^{-1}(C) : C \in gT^{(n)},\ C \subseteq I\} = (gT^{(n)})_I. \quad \square$$

2.2 The recursive coloring

Parameters (fixed for the rest of Section 2). Fix a real Hilbert space H, its closed unit ball B, a map $f : B \to B$, and constants $L > 0$, $\delta > 0$ such that

$$\|f(x) - f(y)\| \le L\|x - y\|, \qquad \|f(x) - x\| \ge \delta \quad (x, y \in B); \tag{2.3}$$

these exist by Lemma 1.2. Fix an integer $D \ge 2$ with

$$D > \frac{4L^2}{\delta^2}. \tag{2.4}$$

Fix internal basic intervals $I_1 < \cdots < I_D$ (the *parents*). They exist: take r with $2^r > 2D$ and $I_j = [(2j-1)2^{-r}, 2j2^{-r}]$. Then $0 < 2^{-r} \le \min I_j$, $\max I_j \le 2D2^{-r} < 1$, and $\max I_j = 2j2^{-r} < (2j+1)2^{-r} = \min I_{j+1}$.

The color of a partition. Define $\mathrm{col}(T) \in B$ for every basic partition T by

$$\mathrm{col}(T) = \begin{cases} f\left(\dfrac{1}{D}\displaystyle\sum_{j=1}^{D} \mathrm{col}(T_{I_j})\right), & \text{if } T \text{ respects every } I_j, \\ 0, & \text{otherwise.} \end{cases} \tag{2.5}$$

This is well defined, by strong induction on $|T|$. Suppose $\mathrm{col}(T') \in B$ is defined for all basic partitions with $|T'| < |T|$. If T does not respect every I_j, then $\mathrm{col}(T) = 0 \in B$. Otherwise each T_{I_j} is a basic partition with $|T_{I_j}| < |T|$ by Lemma 2.1(b), since I_j is internal. So each $\mathrm{col}(T_{I_j}) \in B$ is defined, their average lies in the convex set B, which is the domain of f, and $\mathrm{col}(T) \in B$.

Example (for the displayed choice of I_j; not used later). The one-cell partition $T^{(0)}$ respects no I_j, so $\mathrm{col}(T^{(0)}) = 0$. Each I_j is a cell of $T^{(r)}$ and $(T^{(r)})_{I_j} = T^{(0)}$, so $\mathrm{col}(T^{(r)}) = f(0)$. Likewise $(T^{(2r)})_{I_j} = T^{(r)}$, so $\mathrm{col}(T^{(2r)}) = f(f(0))$.

Parents and children. Let

$$\mathcal{I} = \{I_i : 1 \le i \le D\} \cup \{I_i \cdot I_j : 1 \le i, j \le D\}.$$

By Lemma 2.1(a) each child $I_i \cdot I_j$ is a basic interval contained in I_i; hence every member of $\mathcal{I}$ is internal. The following pairs in $\mathcal{I}$ are separated:

(P) two distinct parents I_k, I_l, $k \neq l$ (by choice);
(S) two siblings $I_i \cdot I_j$, $I_i \cdot I_l$, $j \neq l$ (the increasing affine map s_{I_i} preserves the relation $<$);
(M) a child $I_i \cdot I_j$ and a parent I_k with $k \neq i$ (because $I_i \cdot I_j \subseteq I_i$ and I_i, I_k are separated).

A child and its own parent are nested, not separated.

Colors as functions on F. Call $(n, g) \in \mathbb{N} \times F$ *admissible* if $gT^{(n)}$ is a basic partition respecting every member of $\mathcal{I}$. For every n, $g \in F$ and $I \in \mathcal{I}$ define

$$X_I^{(n)}(g) = \begin{cases} \mathrm{col}((gT^{(n)})_I), & \text{if } (n,g) \text{ is admissible,} \\ 0, & \text{otherwise,} \end{cases} \qquad \varphi_{I,J}^{(n)}(g) = \langle X_I^{(n)}(g), X_J^{(n)}(g) \rangle. \tag{2.6}$$

So $X_I^{(n)} : F \to B$ and $\varphi_{I,J}^{(n)} : F \to [-1, 1]$ are defined on all of F, the latter by Cauchy–Schwarz.

2.3 The uniform boundary bound

The finite set S. For every ordered pair $I < J$ of members of $\mathcal{I}$, use Lemma 2.3 to choose $h_{I,J} \in F$ mapping I affinely onto I_1 and J affinely onto I_2. Let S be the finite set of these elements. It depends only on the choices of Section 2.2.

Lemma 2.5 (Finite averages under translation)**.** *For any nonempty finite $A \subset F$, any $h \in F$, and any $\psi : F \to [-1, 1]$,*

$$\left| \frac{1}{|A|} \sum_{g \in A} \psi(hg) - \frac{1}{|A|} \sum_{g \in A} \psi(g) \right| \le \frac{|hA \triangle A|}{|A|}. \tag{2.7}$$

Proof. Since $g \mapsto hg$ is a bijection $A \to hA$, $\sum_{g \in A} \psi(hg) = \sum_{g' \in hA} \psi(g')$. In $\sum_{hA} \psi - \sum_A \psi$ the terms with $g' \in hA \cap A$ cancel, leaving $\sum_{hA \setminus A} \psi - \sum_{A \setminus hA} \psi$, of absolute value at most $|hA \setminus A| + |A \setminus hA| = |hA \triangle A|$. □

Proposition 2.6. *For every nonempty finite set $A \subset F$,*

$$\max_{h \in S} \frac{|hA \triangle A|}{|A|} \ge \eta_0 := \frac{\delta^2/L^2 - 4/D}{4(1 - 1/D)} > 0. \tag{2.8}$$

Proof. Plan. Fix A and a level n fine enough for $A \cup SA$. Steps 1–2 show that all separated pairs have the same averaged correlation up to the boundary ratio η. Step 3 is the exact identity $X_{I_i} = f(z_i)$. Steps 4–7 turn this into $\delta^2 \le L^2(4/D + 4(1 - 1/D)\eta)$.

Step 0: setup. Fix a nonempty finite $A \subset F$ and put

$$\eta = \max_{h \in S} \frac{|hA \triangle A|}{|A|}, \qquad SA = \bigcup_{h \in S} hA. \tag{2.9}$$

The set $A \cup SA$ is finite. By Lemma 2.2(b), with $\Omega = A \cup SA$ and $\mathcal{J} = \mathcal{I}$, there is n such that (n, g) is admissible for every $g \in A \cup SA$. *Fix this one n for the rest of the proof* and drop the superscript: $X_I = X_I^{(n)}$, $\varphi_{I,J} = \varphi_{I,J}^{(n)}$. Write

$$\mathbb{E}_A \psi = \frac{1}{|A|} \sum_{g \in A} \psi(g), \qquad \alpha = \mathbb{E}_A \varphi_{I_1, I_2}.$$

The level n, the functions X_I, and the number α depend on A; S, D, L, δ do not. All identities below are at this single level.

Step 1: transports carry colors. *Claim: for $I < J$ in $\mathcal{I}$, $h = h_{I,J}$, and $g \in A$,*

$$X_I(g) = X_{I_1}(hg), \qquad X_J(g) = X_{I_2}(hg), \qquad \text{hence} \qquad \varphi_{I,J}(g) = \varphi_{I_1,I_2}(hg).$$

Since $h \in S$ and $g \in A$, we have $hg \in hA \subseteq SA$. So both (n, g) and (n, hg) are admissible (Step 0). Thus $gT^{(n)}$ is basic and respects I, and $(hg)T^{(n)}$ is basic and respects I_1. Since h maps I affinely onto I_1, Lemma 2.4 gives $((hg)T^{(n)})_{I_1} = (gT^{(n)})_I$. Applying col and using the first case of (2.6) for both g and hg gives $X_{I_1}(hg) = X_I(g)$. The same argument with (J, I_2) gives the second identity. It is irrelevant what h does to the other members of $\mathcal{I}$: admissibility of g and of hg was secured separately in Step 0.

Step 2: all separated correlations are within η of α. *Claim:*

$$|\mathbb{E}_A \langle X_I, X_J \rangle - \alpha| \le \eta \qquad \text{whenever } I, J \in \mathcal{I} \text{ are separated.} \tag{2.10}$$

By symmetry of the real inner product we may assume $I < J$. By Step 1, $\mathbb{E}_A \langle X_I, X_J \rangle = \frac{1}{|A|} \sum_{g \in A} \varphi_{I_1,I_2}(h_{I,J} g)$. Apply Lemma 2.5 with $\psi = \varphi_{I_1,I_2}$, which maps F into $[-1, 1]$, and $h = h_{I,J} \in S$; the right-hand side of (2.7) is at most η by (2.9). Nothing is claimed about nested pairs.

Step 3: the recursion as an identity on A. For $g \in A$ put

$$m(g) = \frac{1}{D} \sum_{k=1}^{D} X_{I_k}(g), \qquad z_i(g) = \frac{1}{D} \sum_{j=1}^{D} X_{I_i \cdot I_j}(g) \quad (1 \le i \le D).$$

These are averages of points of B, hence lie in B. *Claim:*

$$X_{I_i}(g) = f(z_i(g)) \qquad (g \in A,\ 1 \le i \le D). \tag{2.11}$$

Let $T = gT^{(n)}$; it is basic and respects I_i and every $I_i \cdot I_j$ by admissibility. By Lemma 2.1(c), T_{I_i} respects every I_j and $(T_{I_i})_{I_j} = T_{I_i \cdot I_j}$. So the first case of (2.5) applies to T_{I_i}:

$$X_{I_i}(g) = \mathrm{col}(T_{I_i}) = f\Big(\frac{1}{D} \sum_{j=1}^{D} \mathrm{col}\big((T_{I_i})_{I_j}\big)\Big) = f\Big(\frac{1}{D} \sum_{j=1}^{D} \mathrm{col}(T_{I_i \cdot I_j})\Big) = f(z_i(g)).$$

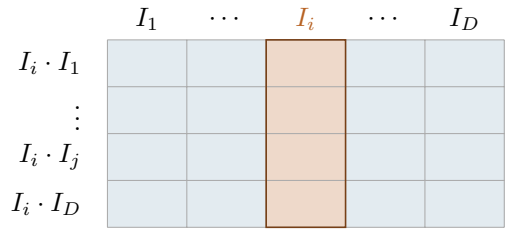


Figure 1: The mixed term for fixed i: the (j,k) entry is $\langle X_{I_i \cdot I_j}, X_{I_k} \rangle$. Rows index the children of I_i, columns index all parents; the grid is schematic. Only column i contains nested pairs, a fraction $1/D$ of the D^2 terms.

Step 4: pointwise displacement inequality. *Claim: for every $g \in A$,*

$$\begin{aligned} \delta^2 &\le \|m(g) - f(m(g))\|^2 \\ &= \left\| \frac{1}{D} \sum_{i=1}^{D} (f(z_i(g)) - f(m(g))) \right\|^2 \\ &\le \frac{1}{D} \sum_{i=1}^{D} \|f(z_i(g)) - f(m(g))\|^2 \\ &\le \frac{L^2}{D} \sum_{i=1}^{D} \|z_i(g) - m(g)\|^2. \end{aligned} \tag{2.12}$$

Line 1 is the displacement bound in (2.3) at the point $m(g) \in B$. Line 2 uses $m(g) = \frac{1}{D}\sum_i f(z_i(g))$, which is (2.11). Line 3 is convexity of the squared norm: $\|\frac{1}{D}\sum_i y_i\|^2 \le (\frac{1}{D}\sum_i \|y_i\|)^2 \le \frac{1}{D}\sum_i \|y_i\|^2$, by the triangle and Cauchy–Schwarz inequalities. Line 4 is the Lipschitz bound in (2.3).

Step 5: three second-moment bounds. *Claim: for each i,*

$$\mathbb{E}_A \|m\|^2 \le \left(1 - \frac{1}{D}\right)(\alpha + \eta) + \frac{1}{D}, \tag{2.13}$$

$$\mathbb{E}_A \|z_i\|^2 \le \left(1 - \frac{1}{D}\right)(\alpha + \eta) + \frac{1}{D}, \tag{2.14}$$

$$\mathbb{E}_A \langle z_i, m \rangle \ge \left(1 - \frac{1}{D}\right)(\alpha - \eta) - \frac{1}{D}. \tag{2.15}$$

For (2.13): by bilinearity and linearity of $\mathbb{E}_A$, $\mathbb{E}_A\|m\|^2 = D^{-2}\sum_{k,l=1}^{D} \mathbb{E}_A\langle X_{I_k}, X_{I_l}\rangle$. The $D(D-1)$ terms with $k \ne l$ are of type (P), so each is at most $\alpha + \eta$ by (2.10). The D terms with $k = l$ are at most 1 because $\|X_{I_k}\| \le 1$. Hence $\mathbb{E}_A\|m\|^2 \le D^{-2}(D(D-1)(\alpha+\eta) + D)$. Each term is bounded individually, so no assumption on the sign of α or $\alpha + \eta$ is needed.

For (2.14): the same computation with the D siblings $I_i \cdot I_1, \ldots, I_i \cdot I_D$; the off-diagonal pairs are of type (S).

For (2.15): $\mathbb{E}_A\langle z_i, m\rangle = D^{-2}\sum_{j,k=1}^{D} \mathbb{E}_A\langle X_{I_i \cdot I_j}, X_{I_k}\rangle$. The $D(D-1)$ terms with $k \ne i$ are of type (M), so each is at least $\alpha - \eta$ by (2.10). The D terms with $k = i$ pair a child with its own parent and are at least -1 by Cauchy–Schwarz. Figure 1 shows this division.

Step 6: the variance estimate. *Claim:*

$$\mathbb{E}_A \|z_i - m\|^2 \le \frac{4}{D} + 4\left(1 - \frac{1}{D}\right)\eta \qquad (1 \le i \le D). \tag{2.16}$$

Expand $\|z_i - m\|^2 = \|z_i\|^2 - 2\langle z_i, m\rangle + \|m\|^2$, average over A, and insert (2.13)–(2.15):

$$\mathbb{E}_A\|z_i - m\|^2 \le 2\Big[\Big(1 - \frac{1}{D}\Big)(\alpha + \eta) + \frac{1}{D}\Big] - 2\Big[\Big(1 - \frac{1}{D}\Big)(\alpha - \eta) - \frac{1}{D}\Big] = 4\Big(1 - \frac{1}{D}\Big)\eta + \frac{4}{D}.$$

The terms in α cancel exactly.

Step 7: conclusion. Average (2.12) over $g \in A$; the left side is the constant δ^2. Then apply (2.16) to each of the D summands:

$$\delta^2 \le \frac{L^2}{D}\sum_{i=1}^{D}\mathbb{E}_A\|z_i - m\|^2 \le L^2\left(\frac{4}{D} + 4\left(1 - \frac{1}{D}\right)\eta\right).$$

Since $D \ge 2$, $1 - 1/D > 0$, and solving for η gives $\eta \ge (\delta^2/L^2 - 4/D)/(4(1 - 1/D)) = \eta_0$. The numerator is positive exactly because $D > 4L^2/\delta^2$, which is (2.4). The constant η_0 does not involve A or n. □

Proof of Theorem 1.1. Suppose F were amenable. Apply the Følner criterion (1.1) to the finite set S of Section 2.3 and $\varepsilon = \eta_0 > 0$ from (2.8). It yields a nonempty finite $A \subset F$ with $|hA \triangle A|/|A| < \eta_0$ for all $h \in S$, contradicting Proposition 2.6. Hence F is not amenable. □

Remark 2.7 (Where each hypothesis enters; commentary, not part of the proof)**.** (a) The group F enters only through Lemma 2.2 and Lemma 2.3. The decisive property is that *one* element moves *two* separated intervals of unrelated sizes and positions affinely onto I_1 and I_2 at once. This is what makes the cross-correlations of types (P), (S), (M) comparable in Step 2, and all three types are needed for the cancellation of α in Step 6. A single affine map cannot do this for all pairs: it multiplies every length by the same factor, so it can carry I onto I_1 and J onto I_2 only if $|I|/|J| = |I_1|/|I_2|$. For the type (M) pair $I = I_1 \cdot I_1 < J = I_2$ we have $|I|/|J| = |I_1|^2/|I_2| \ne |I_1|/|I_2|$, because $|I_1| < 1$. So in a group all of whose elements are affine, such as the solvable (hence amenable) group of maps $x \mapsto 2^k x + b$ of $\mathbb{R}$ with b dyadic, the set S cannot be formed and (2.10) has no analogue for type (M) pairs. This only illustrates where the structure of F is used.

(b) Infinite dimension enters only through Lemma 1.2. If $\dim H < \infty$, every continuous self-map of B has a fixed point (Brouwer), so no map f with $\delta > 0$ exists and the argument cannot start.

(c) The Lipschitz property of f is used only in the last line of (2.12). The inner product is used to form the scalar functions $\varphi_{I,J}$, to which the scalar inequality (2.7) applies.

3 Consequences

This section is not used in the proof of Theorem 1.1. Both corollaries are obtained by inserting Theorem 1.1 into theorems of two companion manuscripts. Those theorems are cited but not stated in the source, so the deductions cannot be checked from the present text; each proof below says exactly what is taken from the companion.

3.1 Uniformly bounded representations

Corollary 3.1. *For every $\varepsilon > 0$, there are a separable complex Hilbert space $\mathcal{H}$ and a representation $\pi : F \to \mathrm{GL}(\mathcal{H})$ such that $\sup_{g\in F}\|\pi(g)\| \le 1 + \varepsilon$, but no bounded invertible operator V on $\mathcal{H}$ makes every $V\pi(g)V^{-1}$ unitary.*

Proof, as given in the source. The group F is countable: an element is determined by a finite list of dyadic breakpoints and integer slope exponents. It is nonamenable by Theorem 1.1. The conclusion is then quoted from the companion unitarizability theorem [17, Theorem 1.1].

Unverified step. The source does not reproduce the statement of [17, Theorem 1.1], and I could not consult it. The deduction is valid only if that theorem gives exactly the displayed conclusion, including the bound $1+\varepsilon$, for every countable nonamenable group. To my knowledge that general statement is the Dixmier unitarizability problem, which was open in the literature I know, so this corollary is conditional on the companion paper.

□

3.2 Percolation on Cayley graphs

Notation for this subsection only. Let $\Sigma \subset F \setminus \{\mathrm{id}\}$ be a finite symmetric generating set and $\mathrm{Cay}(F, \Sigma)$ the simple undirected Cayley graph with vertex set F and edges $\{g, g\sigma\}$, $\sigma \in \Sigma$. In Bernoulli bond percolation each edge is independently open with probability p, under the law $\mathbb{P}_p$. Write p_c for the threshold for existence of an infinite open cluster and p_u for the threshold for almost-sure uniqueness of the infinite open cluster. The source says only "threshold"; the usual conventions are $p_c = \inf\{p : \mathbb{P}_p(\text{there is an infinite open cluster}) > 0\}$ and $p_u = \inf\{p : \mathbb{P}_p(\text{there is exactly one infinite open cluster}) = 1\}$, and the binding definitions are those of the companion result cited below. The connection kernel $\mathcal{C}_p(g, g') = \mathbb{P}_p(g \leftrightarrow g')$ acts on counting-measure $\ell^2(F)$, initially on finitely supported functions by summation over g'. Put $p_{2\to 2} = \sup\{p \in [0,1] : \mathcal{C}_p \text{ is bounded on } \ell^2(F)\}$.

Corollary 3.2. *For every such generating set Σ,*

$$\|\mathcal{C}_{p_c}\|_{2\to 2} < \infty, \qquad p_c < p_{2\to 2} \le p_u.$$

For each fixed $p \in (p_c, p_u)$, there are almost surely infinitely many infinite open clusters. Moreover, there are deterministic numbers $p_c < p_1 < p_2 < 1$, depending on Σ, with the following property. Couple all values of p by giving the edges independent uniform $[0,1]$ labels and declaring an edge open at level p when its label is at most p. Then, with probability one, the level-p configuration has infinitely many infinite open clusters simultaneously for every $p \in [p_1, p_2]$; that is, a single almost-sure event works for all these p (this is how I read the source's word "simultaneously").

Proof, as given in the source. Apply the companion percolation result [16, Corollary 1.2] to F, which is nonamenable by Theorem 1.1.

Unverified step. The source does not reproduce the statement of [16, Corollary 1.2], and I could not consult it. Every assertion of the corollary comes from that result: $\|\mathcal{C}_{p_c}\|_{2\to 2} < \infty$, both inequalities $p_c < p_{2\to 2} \le p_u$, the infinitely many clusters for $p \in (p_c, p_u)$, and the interval $[p_1, p_2]$ with its simultaneity statement. None of them is derived here. To my knowledge $p_c < p_u$ for Cayley graphs of arbitrary nonamenable groups was an open conjecture in the literature I know, so this corollary is conditional on the companion paper.

□

A An explicit map with positive displacement

This appendix proves the Hilbert-space instance of Lemma 1.2 [2] directly. It is independent of Section 2 and its notation is used only here.

Proposition A.1. *Let $H = L^2([0,1];\mathbb{R}^2)$, a real Hilbert space, and $B = \{x \in H : \|x\| \le 1\}$. There is a globally Lipschitz map $f : B \to B$ such that*

$$\|f(x)\| = 1, \qquad \|f(x) - x\| \ge \frac{1}{2} \quad (x \in B).$$

Plan. We construct a Lipschitz map $\Phi : B \to H$ with $\inf_B \|\Phi\| > 0$ and $\Phi(x) = x$ whenever $\|x\| \ge 1/2$; then $f = -\Phi/\|\Phi\|$ works. The identity map fails only because it vanishes at 0. We modify it inside a thin tube around a curve $\gamma(t) = a(t)v(t)$ through 0, with $v(t)$ a unit vector: along the core of the tube the coefficient $a(t)$ of $v(t)$ is pushed down to at most $-1/8$, for all parameters $t \ge -1$. This requires a curve that stays in the ball of radius $5/16$ for all $t \ge 1$ yet never comes close to itself. Such a curve exists in infinite dimensions (Lemma A.3). The steps are: calculus of γ (Lemma A.2); separation (Lemma A.3); nearest-point projection in a tube (Lemma A.4); rotations aligning the tangent with the radial direction (Lemma A.5); the map Φ.

A.1 The curve

For $\xi = (\xi_1, \xi_2) \in \mathbb{R}^2$ put $\xi^\perp = (-\xi_2, \xi_1)$, a linear isometry with $\xi^\perp \cdot \xi = 0$. Put $\mathrm{cs}(\vartheta) = (\cos\vartheta, \sin\vartheta)$, so that $\mathrm{cs}' = \mathrm{cs}^\perp$ and $\mathrm{cs}'' = -\mathrm{cs}$. Define functions on $\mathbb{R}$ by

$$a(t) = \begin{cases} t/8, & t \le 2, \\ 5/16 - (3-t)^2/16, & 2 < t < 3, \\ 5/16, & t \ge 3, \end{cases} \qquad \theta(t) = \begin{cases} 0, & t \le 1, \\ (t-1)^2/2, & 1 < t < 2, \\ t - 3/2, & t \ge 2, \end{cases}$$

and, for $w \in [0,1]$,

$$v(t)(w) = \mathrm{cs}(\theta(t)w), \qquad \hat{v}(t)(w) = w\ \mathrm{cs}(\theta(t)w)^\perp, \qquad \gamma(t) = a(t)v(t).$$

Direct inspection gives the following facts.

(F1) a is continuous ($a(2) = 1/4$, $a(3) = 5/16$ from both sides) and differentiable, with $a' = 1/8$ on $(-\infty, 2]$, $a'(t) = (3-t)/8$ on $[2,3]$, and $a' = 0$ on $[3,\infty)$. Hence $0 \le a' \le 1/8$, a' is $\frac{1}{8}$-Lipschitz, a is strictly increasing on $(-\infty, 3]$, $a \le 5/16$ everywhere, $a(t) \ge 1/8$ for $t \ge 1$, $a(t) \ge 1/4$ for $t \ge 2$, and $a(t) \le -1/8 \iff t \le -1$.

(F2) θ is continuous ($\theta(2) = 1/2$ from both sides) and differentiable, with $\theta' = 0$ on $(-\infty, 1]$, $\theta'(t) = t - 1$ on $[1,2]$, and $\theta' = 1$ on $[2,\infty)$. Hence $0 \le \theta' \le 1$, θ and θ' are 1-Lipschitz, $\theta = 0$ on $(-\infty, 1]$, θ is strictly increasing on $[1,\infty)$, and $\theta(t) \to \infty$.

(F3) $\|v(t)\| = 1$, $\|\hat{v}(t)\|^2 = \int_0^1 w^2\, dw = 1/3$, $\langle v(t), \hat{v}(t)\rangle = 0$, and $\|\gamma(t)\| = |a(t)|$. For $t \le 1$, $v(t)$ is the constant function $(1,0)$ and $\gamma(t) = (t/8)v(t)$ is a straight line through $\gamma(0) = 0$. For $t \ge 1$, $\|\gamma(t)\| \le 5/16$.

Lemma A.2 (Calculus of the curve)**.** *(i) v and $\hat{v}$ are 1-Lipschitz maps $\mathbb{R} \to H$, and v is differentiable in H with $v'(t) = \theta'(t)\hat{v}(t)$.*

(ii) γ is differentiable in H, with

$$\gamma'(t) = a'(t)v(t) + a(t)\theta'(t)\hat{v}(t), \qquad \|\gamma'(t)\|^2 = a'(t)^2 + \frac{a(t)^2\theta'(t)^2}{3}.$$

(iii) $c := \inf_t \|\gamma'(t)\| = \frac{1}{8}$, and $M_\gamma := \sup_t \|\gamma'(t)\| \le \frac{1}{4}$. Consequently γ is M_γ-Lipschitz.

(iv) γ' is Lipschitz, with constant $K_\gamma \le 1$.

(v) The unit tangent $u(t) = \gamma'(t)/\|\gamma'(t)\|$ is Lipschitz with constant $K_u \le 16$; $\langle u(t), v(t)\rangle = a'(t)/\|\gamma'(t)\| \ge 0$; and $u(t) = v(t)$ for $t \le 1$.

(vi) $\langle v(\tau), v(t)\rangle = \operatorname{sinc}(\theta(t) - \theta(\tau))$, where $\operatorname{sinc}(\omega) = \sin(\omega)/\omega$ for $\omega \ne 0$ and $\operatorname{sinc}(0) = 1$.

Proof. (i) Fix $t, \tau \in \mathbb{R}$ and $w \in [0,1]$, and put $\vartheta_0 = \theta(t)w$, $\vartheta_1 = \theta(\tau)w$. By (F2), $|\vartheta_1 - \vartheta_0| \le |\tau - t|$. Since $|\operatorname{cs}'| = 1$, $|\operatorname{cs}(\vartheta_1) - \operatorname{cs}(\vartheta_0)| \le |\tau - t|$; this bounds both $|v(\tau)(w) - v(t)(w)|$ and $|\hat{v}(\tau)(w) - \hat{v}(t)(w)|$ (as $w \le 1$). Integrating the squares over w gives the two Lipschitz claims. For the derivative, Taylor's formula with $|\operatorname{cs}''| = 1$ gives $|\operatorname{cs}(\vartheta_1) - \operatorname{cs}(\vartheta_0) - (\vartheta_1 - \vartheta_0)\operatorname{cs}(\vartheta_0)^{\perp}| \le (\vartheta_1 - \vartheta_0)^2/2 \le (\tau - t)^2/2$. Since θ' is 1-Lipschitz, $|(\vartheta_1 - \vartheta_0) - (\tau - t)\theta'(t)w| = w|\theta(\tau) - \theta(t) - (\tau - t)\theta'(t)| \le (\tau - t)^2/2$. Adding,

$$\left|v(\tau)(w) - v(t)(w) - (\tau - t)\theta'(t)\hat{v}(t)(w)\right| \le (\tau - t)^2 \qquad \text{for every } w \in [0,1].$$

The bound is uniform in w, so $\|v(\tau) - v(t) - (\tau - t)\theta'(t)\hat{v}(t)\| \le (\tau - t)^2$. This is differentiability in H.

(ii) The product rule for the C^1 scalar a and the differentiable H-valued v gives the formula for γ'. The norm follows from (F3) ($v \perp \hat{v}$, $\|v\| = 1$, $\|\hat{v}\|^2 = 1/3$).

(iii) *Lower bound.* For $t \le 2$, $a'(t) = 1/8$, so $\|\gamma'(t)\|^2 \ge 1/64$, with equality for $t \le 1$ where $\theta' = 0$. For $t \ge 2$, $a(t) \ge 1/4$ and $\theta'(t) = 1$, so $\|\gamma'(t)\|^2 \ge a(t)^2/3 \ge 1/48 > 1/64$; this covers both $(2,3)$ and $[3,\infty)$. *Upper bound.* $a'^2 \le 1/64$. The product $a\theta'$ vanishes for $t \le 1$, and for $t \ge 1$ satisfies $0 \le a\theta' \le 5/16$. So $\|\gamma'\|^2 \le 1/64 + 25/768 = 37/768 < 1/16$. *Lipschitz.* For a unit vector $\zeta \in H$, the scalar function $t \mapsto \langle \gamma(t), \zeta\rangle$ has derivative of absolute value at most M_γ. By the mean value theorem $|\langle \gamma(t) - \gamma(\tau), \zeta\rangle| \le M_\gamma |t - \tau|$; take the supremum over ζ.

(iv) Two general facts are used. First, for a bounded Lipschitz scalar ϕ and a bounded Lipschitz H-valued W, $\operatorname{Lip}(\phi W) \le \operatorname{Lip}(\phi)\sup\|W\| + \sup|\phi|\operatorname{Lip}(W)$. Second, a continuous scalar function on $\mathbb{R}$ that is differentiable off finitely many points with derivative bounded by k in absolute value is k-Lipschitz: apply the mean value theorem on each piece between two given points and add. With $\phi = a'$, $W = v$: $\operatorname{Lip}(a'v) \le \frac{1}{8}\cdot 1 + \frac{1}{8}\cdot 1$. The scalar $a\theta'$ is continuous and vanishes on $(-\infty, 1]$, so the unboundedness of a there is harmless. On each of $(1,2), (2,3), (3,\infty)$ it has derivative $a'\theta' + a\theta''$, with $|a'\theta'| \le 1/8$ and $|a\theta''| \le 5/16$ (as $0 < a \le 5/16$ there and $\theta'' \in \{0,1\}$). By the second fact $\operatorname{Lip}(a\theta') \le 7/16$; also $\sup|a\theta'| \le 5/16$. With $\phi = a\theta'$, $W = \hat{v}$: $\operatorname{Lip}(a\theta'\hat{v}) \le \frac{7}{16}\cdot\frac{1}{\sqrt{3}} + \frac{5}{16}\cdot 1$. The total $\frac{1}{4} + \frac{7}{16\sqrt{3}} + \frac{5}{16}$ is less than 1.

(v) *Normalization inequality.* For $r_0 > 0$ and vectors with $\|x\|, \|y\| \ge r_0$,

$$\Big\|\frac{x}{\|x\|} - \frac{y}{\|y\|}\Big\| \le \frac{\|x - y\|}{\|x\|} + \|y\|\Big|\frac{1}{\|x\|} - \frac{1}{\|y\|}\Big| \le \frac{2\|x - y\|}{r_0}.$$

With $r_0 = c = 1/8$, applied to $\gamma'(t), \gamma'(\tau)$, this gives $K_u \le 16K_\gamma \le 16$. Next, $\langle \gamma', v\rangle = a'\|v\|^2 + a\theta'\langle \hat{v}, v\rangle = a' \ge 0$. For $t \le 1$, $\gamma'(t) = \frac{1}{8}v(t)$, so $u(t) = v(t)$.

(vi) Pointwise, $\operatorname{cs}(\theta(\tau)w)\cdot\operatorname{cs}(\theta(t)w) = \cos((\theta(t) - \theta(\tau))w)$, and $\int_0^1 \cos(\omega w)\,dw = \operatorname{sinc}(\omega)$. □

Lemma A.3 (Injectivity and uniform separation)**.** *The curve γ is injective, and for every $d > 0$,*

$$\sigma(d) := \inf_{|t-\tau|\ge d} \|\gamma(t) - \gamma(\tau)\| > 0. \tag{A.1}$$

This is what permits a tube of one fixed radius along the whole curve in Lemma A.4.

Proof. Injectivity. Let $\tau < t$ and suppose $\gamma(\tau) = \gamma(t)$.

- If $t \le 1$: $v(\tau) = v(t)$ is the same unit vector by (F3), so $a(\tau) = a(t)$, contradicting strict monotonicity of a on $(-\infty, 1]$.

- If $t > 1$: by (F2), $\theta(\tau) < \theta(t)$ (whether $\tau \le 1$ or $\tau > 1$). By Lemma A.2(vi) and $|\operatorname{sinc}(\omega)| < 1$ for $\omega \neq 0$, $|\langle v(\tau), v(t)\rangle| < 1$, so the unit vectors $v(\tau), v(t)$ are linearly independent. Then $a(\tau)v(\tau) = a(t)v(t)$ forces $a(t) = 0$, contradicting $a(t) \ge 1/8$ for $t \ge 1$. This covers every sign of $a(\tau)$ and the range $t, \tau \ge 3$ where a is constant.

Separation. Suppose (A.1) fails for some d. Choose $\tau_n < t_n$ with $t_n - \tau_n \ge d$ and $\|\gamma(t_n) - \gamma(\tau_n)\| \to 0$, and pass to a subsequence along which $\tau_n \to \tau_\infty$ and $t_n \to t_\infty$ in $[-\infty, \infty]$. Since $\tau_n < t_n$, the possibilities are as follows.

- *Both limits finite.* Then $t_\infty - \tau_\infty \ge d$ and, by continuity, $\gamma(t_\infty) = \gamma(\tau_\infty)$, contradicting injectivity.
- *Both $-\infty$.* Eventually $t_n, \tau_n \le 1$, and then $\|\gamma(t_n) - \gamma(\tau_n)\| = (t_n - \tau_n)/8 \ge d/8$.
- $\tau_\infty = -\infty$, $t_\infty > -\infty$. Then $\|\gamma(\tau_n)\| = |a(\tau_n)| \to \infty$ while $\|\gamma(t_n)\| = |a(t_n)|$ stays bounded.
- τ_∞ *finite,* $t_\infty = +\infty$. Then $\theta(t_n) - \theta(\tau_n) \to \infty$, so $\langle v(\tau_n), v(t_n)\rangle \to 0$ by Lemma A.2(vi), and $\|\gamma(t_n) - \gamma(\tau_n)\|^2 = a(t_n)^2 + a(\tau_n)^2 - 2a(t_n)a(\tau_n)\langle v(\tau_n), v(t_n)\rangle \to (5/16)^2 + a(\tau_\infty)^2 > 0$.
- *Both $+\infty$.* Eventually $t_n, \tau_n \ge 3$, so $a = 5/16$ and $\theta(t_n) - \theta(\tau_n) = t_n - \tau_n \ge d$, whence $\|\gamma(t_n) - \gamma(\tau_n)\|^2 = 2(5/16)^2(1 - \operatorname{sinc}(t_n - \tau_n))$. This is bounded away from 0 because $\sup_{\omega \ge d} \operatorname{sinc}(\omega) < 1$: sinc is continuous, less than 1 on $(0, \infty)$, and tends to 0 at infinity.

Each case contradicts $\|\gamma(t_n) - \gamma(\tau_n)\| \to 0$. □

Lemma A.4 (Tube and nearest-point projection)**.** *There is a radius $0 < \rho < 1/32$ such that every point of*

$$U = \{x \in H : \operatorname{dist}(x, \gamma(\mathbb{R})) < \rho\}$$

has a unique nearest point $\gamma(t(x))$ on the curve. Writing $e(x) = x - \gamma(t(x))$, we have $\|e(x)\| = \operatorname{dist}(x, \gamma(\mathbb{R})) < \rho$, $e(x) \perp u(t(x))$, and

$$|t(x) - t(y)| \le 16\|x - y\| \quad \text{if } x, y \in U \text{ and } \|x - y\| < \rho. \tag{A.2}$$

Proof. Step 1 (tangent estimate). Choose $d > 0$ with $K_\gamma d \le c/2 = 1/16$; $d = 1/16$ works by Lemma A.2(iv). *Claim:*

$$|\langle \gamma(t) - \gamma(\tau), u(\tau)\rangle| \ge \frac{3c}{4}|t - \tau| = \frac{3}{32}|t - \tau| \qquad (|t - \tau| \le d). \tag{A.3}$$

Fix τ and let $\psi(t) = \langle \gamma(t), u(\tau)\rangle$. Then $\psi'(t) = \langle \gamma'(t), u(\tau)\rangle$ is K_γ-Lipschitz and $\psi'(\tau) = \|\gamma'(\tau)\| \ge c$. Hence $|\psi(t) - \psi(\tau) - (t - \tau)\psi'(\tau)| = |\int_\tau^t (\psi'(\varsigma) - \psi'(\tau))\, d\varsigma| \le \frac{K_\gamma}{2}|t - \tau|^2 \le \frac{c}{4}|t - \tau|$ when $|t - \tau| \le d$, and (A.3) follows.

Step 2 (choice of ρ). With $\sigma(d) > 0$ from Lemma A.3, choose $\rho > 0$ such that

$$\rho < \frac{1}{32}, \qquad 4\rho < \sigma(d), \qquad \rho K_u < \frac{c}{4} = \frac{1}{32}. \tag{A.4}$$

Step 3 (a nearest point exists and $e \perp u$). Let $x \in U$ and take t_n with $\|x - \gamma(t_n)\| \to \operatorname{dist}(x, \gamma(\mathbb{R})) < \rho$. Choose N with $\|x - \gamma(t_n)\| < 2\rho$ for all $n \ge N$. For $n \ge N$, $\|\gamma(t_n) - \gamma(t_N)\| < 4\rho < \sigma(d)$. Since $\sigma(d)$ is the infimum of such distances over parameters differing by at least d, this forces $|t_n - t_N| < d$. So $t_n \in [t_N - d, t_N + d]$ for all $n \ge N$, a compact interval. A subsequence converges to some t_*, and by continuity $\gamma(t_*)$ attains the distance. At any minimizing parameter t, the derivative of $t \mapsto \|x - \gamma(t)\|^2$, namely $-2\langle x - \gamma(t), \gamma'(t)\rangle$, vanishes; hence $(x - \gamma(t)) \perp u(t)$.

Step 4 (parameter estimate and uniqueness). Let $x, y \in U$ with $\|x - y\| < \rho$. Uniqueness is not yet known, so let t be *any* minimizing parameter for x and τ any for y, and write $e_t = x - \gamma(t)$,

$e_\tau = y - \gamma(\tau)$ for the corresponding residuals (once uniqueness is proved these are $e(x)$, $e(y)$). By Step 3, $\|e_t\|, \|e_\tau\| < \rho$, $e_t \perp u(t)$ and $e_\tau \perp u(\tau)$. Then $\|\gamma(t) - \gamma(\tau)\| < 3\rho < \sigma(d)$, so $|t - \tau| < d$. Using the two orthogonality relations,

$$\langle x - y, u(\tau)\rangle = \langle \gamma(t) - \gamma(\tau), u(\tau)\rangle + \langle e_t, u(\tau) - u(t)\rangle.$$

By (A.3), $\|e_t\| < \rho$, and (A.4),

$$\|x - y\| \geq |\langle x - y, u(\tau)\rangle| \geq \Big(\frac{3c}{4} - \rho K_u\Big)|t - \tau| \geq \frac{c}{2}|t - \tau| = \frac{|t - \tau|}{16}.$$

Taking $y = x$ shows that the minimizing parameter is unique. Then the same inequality is (A.2). □

Lemma A.5 (Rotations)**.** *There is a family of orthogonal operators $R_t : H \to H$, $t \in \mathbb{R}$, Lipschitz in operator norm with some constant K_R, such that $R_t u(t) = v(t)$ for all t and $R_t = \mathrm{Id}$ for $t \leq 1$.*

It is used so that a residual $e \perp u(t)$ is carried to a vector perpendicular to $v(t)$. Then changing the coefficient along $v(t)$ cannot be cancelled by the residual.

Proof. Fix t, write $u = u(t)$, $v = v(t)$, and $\kappa = \langle u, v\rangle \geq 0$ (Lemma A.2(v)). Define, for $\zeta \in H$,

$$\Lambda\zeta = v\langle u, \zeta\rangle - u\langle v, \zeta\rangle, \qquad R = \mathrm{Id} + \Lambda + \frac{\Lambda^2}{1 + \kappa}.$$

Case u, v linearly independent. Λ is skew-adjoint, vanishes on $\{u, v\}^\perp$, and maps the plane $\Pi = \mathrm{span}\{u, v\}$ into itself, with $\Lambda u = v - \kappa u$ and $\Lambda v = \kappa v - u$. Hence $\Lambda^2 u = \Lambda v - \kappa\Lambda u = -(1 - \kappa^2)u$ and $\Lambda^2 v = \kappa\Lambda v - \Lambda u = -(1 - \kappa^2)v$, that is, $\Lambda^2 = -(1 - \kappa^2)\mathrm{Id}$ on Π. So on Π, $R = (1 - \frac{1-\kappa^2}{1+\kappa})\mathrm{Id} + \Lambda = \kappa\,\mathrm{Id} + \Lambda$. This restriction satisfies $(\kappa\mathrm{Id} + \Lambda)^*(\kappa\mathrm{Id} + \Lambda) = \kappa^2\mathrm{Id} - \Lambda^2 = \mathrm{Id}$ on Π, so it is orthogonal there, and $Ru = \kappa u + v - \kappa u = v$. On $\Pi^\perp$, $R = \mathrm{Id}$. Thus R is orthogonal on H.

Case u, v linearly dependent. Unit vectors with $\langle u, v\rangle \geq 0$ that are dependent are equal, so $\Lambda = 0$, $R = \mathrm{Id}$, and $Ru = v$. In particular $R_t = \mathrm{Id}$ for $t \leq 1$.

Lipschitz dependence. Fix parameters t, τ and index the objects by them: $u_t, v_t, \kappa_t, \Lambda_t$ and $u_\tau, v_\tau, \kappa_\tau, \Lambda_\tau$. Put $\Delta = \|u_t - u_\tau\| + \|v_t - v_\tau\| \leq (K_u + 1)|t - \tau|$, by Lemma A.2(v),(i).

- Λ: for $\zeta \in H$, $\Lambda_t\zeta - \Lambda_\tau\zeta = (v_t - v_\tau)\langle u_t, \zeta\rangle + v_\tau\langle u_t - u_\tau, \zeta\rangle - (u_t - u_\tau)\langle v_t, \zeta\rangle - u_\tau\langle v_t - v_\tau, \zeta\rangle$, so $\|\Lambda_t - \Lambda_\tau\| \leq 2\Delta$. Also $\|\Lambda_t\| \leq 2$.
- Λ^2: $\Lambda_t^2 - \Lambda_\tau^2 = \Lambda_t(\Lambda_t - \Lambda_\tau) + (\Lambda_t - \Lambda_\tau)\Lambda_\tau$, so $\|\Lambda_t^2 - \Lambda_\tau^2\| \leq 4\|\Lambda_t - \Lambda_\tau\| \leq 8\Delta$. Also $\|\Lambda_\tau^2\| \leq 4$.
- κ: $|\kappa_t - \kappa_\tau| = |\langle u_t - u_\tau, v_t\rangle + \langle u_\tau, v_t - v_\tau\rangle| \leq \Delta$. Since $\kappa_t, \kappa_\tau \geq 0$, $|\frac{1}{1+\kappa_t} - \frac{1}{1+\kappa_\tau}| = \frac{|\kappa_\tau - \kappa_t|}{(1+\kappa_t)(1+\kappa_\tau)} \leq \Delta$, and $0 < \frac{1}{1+\kappa_t} \leq 1$.
- $\Lambda^2/(1 + \kappa)$: $\frac{\Lambda_t^2}{1+\kappa_t} - \frac{\Lambda_\tau^2}{1+\kappa_\tau} = \frac{\Lambda_t^2 - \Lambda_\tau^2}{1+\kappa_t} + \Lambda_\tau^2(\frac{1}{1+\kappa_t} - \frac{1}{1+\kappa_\tau})$, of norm at most $8\Delta + 4\Delta$.

Adding the contributions of Λ and of $\Lambda^2/(1 + \kappa)$, $\|R_t - R_\tau\| \leq (2 + 8 + 4)\Delta \leq 14(K_u + 1)|t - \tau|$, so $K_R \leq 14(K_u + 1)$. The formula for R is the same in both cases, so this holds also where $u = v$. □

A.2 Proof of Proposition A.1

Fix ρ, U, $t(\cdot)$, $e(\cdot)$ from Lemma A.4 and R_t from Lemma A.5.

Step 0: definition of Φ. Set

$$b(t) = \min\{a(t), -1/8\}, \qquad q(t) = b(t) - a(t) = -\max\{a(t) + 1/8, 0\}.$$

By (F1): $|q| \le 5/16 + 1/8 = 7/16$, $\mathrm{Lip}(q) \le \mathrm{Lip}(a) = 1/8$, and $q(t) = 0$ for $t \le -1$. Let $\beta, \chi : [0,\infty) \to [0,1]$ be

$$\beta(s) = \begin{cases} 1, & s \le \rho/4, \\ 2 - 4s/\rho, & \rho/4 < s < \rho/2, \\ 0, & s \ge \rho/2, \end{cases} \qquad \chi(s) = \begin{cases} 1, & s \le \rho/2, \\ 2 - 2s/\rho, & \rho/2 < s < \rho, \\ 0, & s \ge \rho, \end{cases}$$

with $\mathrm{Lip}(\beta) = 4/\rho$, $\mathrm{Lip}(\chi) = 2/\rho$, and $\beta \le \chi$. Define $\Phi : B \to H$ by $\Phi(x) = x$ for $x \in B \setminus U$, and, for $x \in B \cap U$ with $t = t(x)$, $e = e(x)$,

$$\begin{aligned} \Phi(x) &= \big(a(t) + \beta(\|e\|)q(t)\big)v(t) + \big((1 - \chi(\|e\|))\mathrm{Id} + \chi(\|e\|)R_t\big)e \\ &= x + \beta(\|e\|)q(t)v(t) + \chi(\|e\|)(R_t - \mathrm{Id})e, \end{aligned} \tag{A.5}$$

the two lines being equal because $x = a(t)v(t) + e$.

Step 1: Φ is Lipschitz on B. We treat four kinds of pairs $x, y \in B$.

(1a) $x, y \in U$, $\|x - y\| < \rho$. Write t, e for x and τ, e' for y, and $\Delta = \|x - y\|$. From Lemma A.4 and Lemma A.2(iii):

$$|t - \tau| \le 16\Delta, \qquad \|e - e'\| \le \Delta + \|\gamma(t) - \gamma(\tau)\| \le (1 + 16M_\gamma)\Delta, \qquad \big|\|e\| - \|e'\|\big| \le \Delta,$$

the last because $\|e(\cdot)\| = \mathrm{dist}(\cdot, \gamma(\mathbb{R}))$ is 1-Lipschitz. By the second line of (A.5), $\Phi(x) - \Phi(y)$ is $x - y$ plus the differences of the two correction terms. For a product of three bounded Lipschitz factors, the difference is at most the sum over factors of (variation of that factor) $\times$ (bounds of the other two).

- $\beta(\|e\|)\, q(t)\, v(t)$: bounds 1, 7/16, 1; variations $\frac{4}{\rho}\Delta$, $\frac{1}{8} \cdot 16\Delta$, 16Δ (Lemma A.2(i)). Difference at most $\big(\frac{7}{4\rho} + 2 + 7\big)\Delta$.
- $\chi(\|e\|)\,(R_t - \mathrm{Id})\, e$: bounds 1, 2 (as $\|R_t\| = 1$), ρ; variations $\frac{2}{\rho}\Delta$, $16K_R\Delta$, $(1 + 16M_\gamma)\Delta$. Difference at most $\big(4 + 16K_R\rho + 2 + 32M_\gamma\big)\Delta$.

Adding Δ for the term $x - y$ and the two bounds $9 + \frac{7}{4\rho}$ and $6 + 16K_R\rho + 32M_\gamma$: $\|\Phi(x) - \Phi(y)\| \le C_1\|x - y\|$ with $C_1 = 1 + 9 + 6 + \frac{7}{4\rho} + 16K_R\rho + 32M_\gamma = 16 + \frac{7}{4\rho} + 16K_R\rho + 32M_\gamma$.

(1b) $\|x - y\| \ge \rho$. For $x \in B \cap U$, (A.5) gives $\|\Phi(x)\| \le 1 + \frac{7}{16} + 2\rho < \frac{3}{2}$; outside U, $\|\Phi(x)\| \le 1$. So $\|\Phi(x) - \Phi(y)\| \le 3 \le \frac{3}{\rho}\|x - y\|$.

(1c) $x \in U$, $y \notin U$, $\|x - y\| < \rho$. For $0 \le s < \rho$ we have $\beta(s) \le \chi(s) \le 2(\rho - s)/\rho$: with equality on $[\rho/2, \rho)$, and because $\rho - s \ge \rho/2$ on $[0, \rho/2]$. By the second line of (A.5),

$$\|\Phi(x) - x\| \le \frac{2(\rho - \|e\|)}{\rho}\Big(\frac{7}{16} + 2\rho\Big) = \Big(\frac{7}{8\rho} + 4\Big)(\rho - \|e\|).$$

Since $\mathrm{dist}(y, \gamma(\mathbb{R})) \ge \rho$ and the distance function is 1-Lipschitz, $\rho - \|e\| \le \|x - y\|$. With $\Phi(y) = y$, $\|\Phi(x) - \Phi(y)\| \le \|\Phi(x) - x\| + \|x - y\| \le (\frac{7}{8\rho} + 5)\|x - y\|$.

(1d) $x, y \notin U$. Φ is the identity.

Up to exchanging x and y, these cases are exhaustive, so Φ is Lipschitz with constant $C_\Phi = \max\{C_1, \frac{3}{\rho}, \frac{7}{8\rho} + 5, 1\}$.

Step 2: $\|\Phi(x)\| \geq \rho/4$ for all $x \in B$. *Outside U:* since $\gamma(0) = 0$, $\|x\| \geq \operatorname{dist}(x, \gamma(\mathbb{R})) \geq \rho$. *Inside U*, use the first line of (A.5) and note that $e \perp u(t)$ implies $R_t e \perp R_t u(t) = v(t)$ and $\|R_t e\| = \|e\|$.

- $\|e\| \leq \rho/4$: $\beta = \chi = 1$, so $\Phi(x) = b(t)v(t) + R_t e$ with orthogonal summands, and $\|\Phi(x)\| \geq |b(t)| \geq 1/8 > \rho/4$.
- $\rho/4 < \|e\| \leq \rho/2$: $\chi = 1$, so $\Phi(x)$ is a multiple of $v(t)$ plus $R_t e \perp v(t)$, and $\|\Phi(x)\| \geq \|e\| > \rho/4$.
- $\rho/2 < \|e\| < \rho$: $\beta = 0$, so $\Phi(x) = a(t)v(t) + e_*$ with $e_* = ((1-\chi)\mathrm{Id} + \chi R_t)e$ and $\|e_*\| \leq \|e\| < \rho$. If $t > 1$, then $a(t) \geq 1/8$ and $\|\Phi(x)\| \geq 1/8 - \rho > 3/32 > \rho/4$. If $t \leq 1$, then $R_t = \mathrm{Id}$, $e_* = e \perp u(t) = v(t)$, and $\|\Phi(x)\| \geq \|e\| > \rho/2$.

Step 3: $\Phi(x) = x$ whenever $\|x\| \geq 1/2$. Outside U this is the definition. Inside U, $|a(t)| = \|\gamma(t)\| \geq \|x\| - \|e\| > \frac{1}{2} - \rho > \frac{15}{32} > \frac{5}{16}$. Since $a \leq 5/16$ everywhere, $a(t) < -5/16$. Hence $t \leq -1$, so $q(t) = 0$ and $R_t = \mathrm{Id}$, and the second line of (A.5) gives $\Phi(x) = x$.

Step 4: the map f. Define $f(x) = -\Phi(x)/\|\Phi(x)\|$ for $x \in B$; this is legitimate by Step 2, and $\|f(x)\| = 1$. By the normalization inequality in the proof of Lemma A.2(v) with $r_0 = \rho/4$, $y \mapsto y/\|y\|$ is $(8/\rho)$-Lipschitz on $\{\|y\| \geq \rho/4\}$. With Step 1, f is Lipschitz with constant $8C_\Phi/\rho$. Displacement:

- if $\|x\| < 1/2$, then $\|f(x) - x\| \geq \|f(x)\| - \|x\| > 1/2$;
- if $\|x\| \geq 1/2$, then $\Phi(x) = x$ by Step 3, so $f(x) = -x/\|x\|$ and $\|f(x) - x\| = \|x\|\,(1 + 1/\|x\|) = 1 + \|x\| \geq 3/2$.

This proves Proposition A.1, and hence Lemma 1.2 with $\delta = 1/2$. □

Verification checklist

A sceptical reader should check these load-bearing points first.

1. **Exact covariance and its use** (Lemma 2.4; Step 1 of Proposition 2.6). Check that $X_I(g) = X_{I_1}(h_{I,J}g)$ is an exact equality of colors for every $g \in A$. It needs $hg = h \circ g$, $h \circ s_I = s_{I_1}$, and admissibility of both g and $h_{I,J}g$ at the *same* level n (Step 0, via Lemma 2.2(b) applied to $A \cup SA$).
2. **Pair transport** (Lemma 2.3). Check that one element of F moves both intervals affinely, and that all slopes are powers of 2 after the bisections.
3. **Order of quantifiers** (Section 1.3, "Order of choices"; Step 0). S and η_0 are fixed before A; only n and α depend on A; the final bound contains neither.
4. **The recursion identity** $X_{I_i} = f(z_i)$ (Step 3). It rests on $(T_{I_i})_{I_j} = T_{I_i \cdot I_j}$ (Lemma 2.1(c)) and on well-foundedness of (2.5) (Lemma 2.1(b)).
5. **The variance computation** (Steps 5–6). Check which pairs are separated (types (P), (S), (M) in Section 2.2), that exactly D of the D^2 terms in each sum are not, and that α cancels without any sign assumption.
6. **The final inequality** (Steps 4 and 7). Line 2 of (2.12) uses $m = \frac{1}{D}\sum_i f(z_i)$; positivity of η_0 uses $D > 4L^2/\delta^2$.
7. **Existence of f** (Appendix A). The delicate points are the uniform separation (A.1) (Lemma A.3, five limit cases), the lower bound $\|\Phi\| \geq \rho/4$ (Step 2 of the proof of Proposition A.1), and the Lipschitz estimate across the tube boundary (Step 1c there). Alternatively, take Lemma 1.2 from [2].

8. **The Følner implication** (1.1) is the only cited fact about amenability; it is quoted, not proved.
9. **The corollaries** (Section 3) rest on companion theorems not stated in the source; see the two boxed notes there.

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